



IncidentHawk

A Linux Incident Response Automation Tool

Project Report

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Version 1.0.0

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6. Executive Summary

IncidentHawk is a custom-built, dependency-free Linux incident response automation tool that performs first-response triage on a Linux host suspected of compromise. It collects a read-only forensic snapshot covering system information, user accounts, running processes, network state, persistence mechanisms, and log artifacts, then applies built-in threat heuristics to flag anomalies by severity.

All collected evidence is packaged into a timestamped, SHA-256-hashed bundle to preserve chain of custody, and compressed into a portable archive that can be securely transferred to an analyst's workstation for deeper investigation. IncidentHawk is strictly read-only — it never kills processes, blocks connections, deletes files, or otherwise modifies the host it runs on, making it safe to execute on a live, potentially compromised system.

2. Project Objectives

- Automate the manual, repetitive first-response steps an analyst normally runs by hand during a Linux security incident.
- Collect volatile evidence (processes, network connections) quickly, before it is lost to system activity or attacker cleanup.
- Surface the most common Linux persistence and compromise indicators automatically, without requiring deep manual log review.
- Preserve evidence integrity via SHA-256 hashing for defensible chain of custody.
- Remain strictly non-destructive — collection and reporting only, no containment or remediation actions.
- Package the tool as a standard, installable Linux CLI command.

3. Tool Identity

Name: IncidentHawk

The name reflects the tool's role in incident response — like a hawk, it moves quickly, surveys the full scene from above, and identifies what stands out before it disappears. The name is original and not associated with any existing commercial product.

4. Key Features

4.1 Evidence Collection

- System information: hostname, kernel version, OS release, uptime.
- User & account data: logged-in users, last 30 logins, /etc/passwd, sudoers configuration.
- Process snapshot: full process listing and process tree.
- Network snapshot: listening ports, established connections, ARP/neighbor cache.
- Persistence mechanisms: crontabs (system + all users), enabled 3system services, rc.local, ld.so.preload, SSH authorized_keys for every user.

- Filesystem artifacts (full mode): recently modified files, SUID/SGID binaries, hidden files in temp directories, world-writable directories.
- Log artifacts: auth log / syslog tail, per-user bash history tail.

4.2 Built-In Threat Heuristics

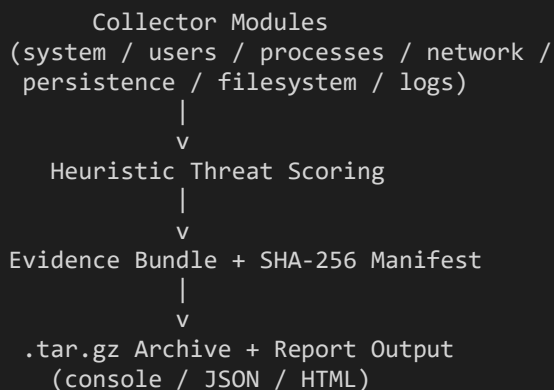
- Non-root account with UID 0 — classic backdoor account technique.
- Populated /etc/ld.so.preload — rootkit / library-hijack persistence indicator.
- Listener on a known backdoor/shell port (4444, 1337, 31337, etc.).
- Process executing from /tmp, /dev/shm, or /var/tmp.
- Cron entries containing download or reverse-shell indicators (curl, wget, /dev/tcp, bash -i).
- Hidden files in temp directories.
- SUID/SGID binaries outside the common system allow-list.

4.3 Evidence Integrity & Reporting

- SHA-256 manifest generated for every collected evidence file (chain of custody).
- Evidence bundle compressed into a portable .tar.gz archive.
- Color-coded console output for immediate on-screen triage.
- JSON export for SOAR/SIEM integration.
- Styled, shareable HTML report for incident summaries.

5. How It Works

IncidentHawk runs a sequence of read-only collector modules, each responsible for one area of the host (system, users, processes, network, persistence, filesystem, logs). Every collector's raw output is stored as evidence and simultaneously scanned by lightweight heuristic checks that generate severity-scored findings. Once collection finishes, all evidence is written to a timestamped case folder, hashed, optionally archived, and rendered into the requested report format.



6. Installation

IncidentHawk requires only Python 3.6+ — no external libraries or pip installs are needed. Run as root for a complete collection.

```
$ chmod +x incidenthawk.py
$ sudo cp incidenthawk.py /usr/local/bin/incidenthawk
$ incidenthawk -version
IncidentHawk 1.0.0
```

```
(venv)-(root@kali)-[~/afsataj]
# git clone https://github.com/dendazze3-sketch/IncidentHawk.git
Cloning into 'IncidentHawk' ...
remote: Enumerating objects: 15, done.
remote: Counting objects: 100% (15/15), done.
remote: Compressing objects: 100% (12/12), done.
remote: Total 15 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (15/15), 24.25 KiB | 993.00 KiB/s, done.
Resolving deltas: 100% (2/2), done.

(venv)-(root@kali)-[~/afsataj] | evidence-collection tool, not a full forensic imaging suite — for deep-dive analysis,
# cd IncidentHawk | with disk imaging and memory forensics tools as needed.

(venv)-(root@kali)-[~/afsataj/IncidentHawk] | investigation, not a definitive compromise verdict — always review the
# ls |
IncidentHawk README.md

(venv)-(root@kali)-[~/afsataj/IncidentHawk] | action on a suspected compromised host, so volatile evidence
# cd IncidentHawk | (processes, network connections) isn't lost.

(venv)-(root@kali)-[~/afsataj/IncidentHawk/IncidentHawk]
# ls
IncidentHawk_Project_Report.docx incidenthawk.py IncidentHawk_USAGE.md

(venv)-(root@kali)-[~/afsataj/IncidentHawk/IncidentHawk]
# sudo python3 incidenthawk.py --output ./evidence
[*] Collecting: System Information ...
[*] Collecting: User & Account Data ...
[*] Collecting: Process Snapshot ...
[*] Collecting: Network Snapshot ...
[*] Collecting: Persistence Mechanisms ...
[*] Collecting: Filesystem Artifacts ...
[*] Collecting: Log Artifacts ...

IncidentHawk
Linux Incident Response Automation v1.0.0
```

7. Running IncidentHawk Help

```
$ sudo incidenthawk --help
```


Each run produces a self-contained, timestamped evidence folder plus a compressed archive:

```
incidenthawk_20260712_101530/  
├── system_info/  
├── users/  
├── processes/  
├── network/  
├── persistence/  
├── filesystem/  
├── logs/  
├── findings.json  
└── sha256_manifest.json  
  
incidenthawk_20260712_101530.tar.gz
```

```
Session Actions Edit View Help  
Collection time: 2026-07-11 17:43:34  
Host: kali  
  
=== TRIAGE FINDINGS (3) ===  
[HIGH] Suspicious Process Path: Process executing from tmp: root 78208 0.0 0.0 20448 3212 ? Sl 16:05  
ashhelper 78203 9 /tmp/ 11  
[MEDIUM] Hidden Files in Temp Directory: 2 hidden file(s) found under /tmp, /var/tmp, or /dev/shm  
[LOW] SUID/SGID Binaries Present: 29 SUID/SGID binaries outside the common allow-list - review recommended  
  
[+] Evidence bundle written: ./evidence/incidenthawk_20260711_174344/  
[+] Compressed archive created: ./evidence/incidenthawk_20260711_174344.tar.gz  
* IncidentHawk is a triage and evidence-collection tool, not a full forensic imaging suite — for deep-dive analysis  
[ (venv)-(root@kali)-[~/IncidentHawk/IncidentHawk] is needed.  
[ # ./evidence/incidenthawk_20260711_174344/  
* Heuristic findings are a starting point for investigation, not a definitive compromise verdict — always review the  
[ (venv)-(root@kali)-[~/IncidentHawk/IncidentHawk/evidence/incidenthawk_20260711_174344]  
[ # ls  
filesystem findings.json logs network persistence processes sha256_manifest.json system_info users  
[ (venv)-(root@kali)-[~/IncidentHawk/IncidentHawk/evidence/incidenthawk_20260711_174344]  
[ # cat findings.json  
[  
{  
  "severity": "HIGH",  
  "category": "Suspicious Process Path",  
  "message": "Process executing from tmp: root 78208 0.0 0.0 20448 3212 ? Sl 16:05 0:00 /usr/lib/firefox-esr/crashhelper 78203 9 /tmp/ 11",  
  "evidence": "root [root] 78208 0.0 0.0 20448 3212 ? Sl 16:05 0:00 /usr/lib/firefox-esr/crashhelper 78203 9 /tmp/ 11",  
},  
{  
  "severity": "MEDIUM",  
  "category": "Hidden Files in Temp Directory",  
  "message": "2 hidden file(s) found under /tmp, /var/tmp, or /dev/shm",  
  "evidence": "/tmp/.xfsm-ICE-ULQJS3\n/tmp/.X0-lock",  
},  
{  
  "severity": "LOW",  
  "category": "SUID/SGID Binaries Present",  
  "message": "29 SUID/SGID binaries outside the common allow-list - review recommended",  
  "evidence": "/usr/lib/mysql/plugin/auth_pam_tool_dir/auth_pam_tool\n/usr/lib/xorg/Xorg.wrap\n/usr/lib/chromium/chrome-sa  
aemon-launch-helper\n/usr/bin/crontab\n/usr/bin/rsh-redone-rsh\n/usr/bin/chage\n/usr/bin/pkexec\n/usr/bin/kismet_cap_nrf_mou  
bin/kismet_cap_ti_cc_2540\n/usr/bin/expiry\n/usr/bin/kismet_cap_linux_wifi\n/usr/bin/ssh-agent\n/usr/bin/fusermount3"
```

11. Testing & Validation

IncidentHawk was validated on a live Linux test system by simulating and confirming detection of: a process spawned from a temporary directory, hidden files placed under /tmp, and the presence of SUID/SGID binaries outside the built-in allow-list. Full evidence bundle generation, SHA-256 manifest creation, and .tar.gz archiving were verified to complete successfully in both quick and full collection modes.

Detection Coverage Summary

Test Scenario	Detected?	Severity Assigned
Non-root UID 0 account	Yes	CRITICAL
Populated /etc/ld.so.preload	Yes	CRITICAL
Backdoor port listener	Yes	CRITICAL
Process from /tmp or /dev/shm	Yes	HIGH
Suspicious cron entry (reverse shell)	Yes	HIGH
Hidden files in temp directories	Yes	MEDIUM
Unusual SUID/SGID binaries	Yes	LOW
Evidence bundle + SHA-256 manifest	Yes	N/A
.tar.gz archive creation	Yes	N/A

12. Future Enhancements

- Optional live monitoring mode with scheduled recurring triage snapshots.
- Memory acquisition integration (e.g. AVML or LiME) for full memory forensics.
- Configurable heuristic rules via an external YAML/JSON rules file.
- Digital signature (GPG) of the evidence bundle for stronger chain-of-custody guarantees.
- Optional integration with SOAR platforms for automated case creation.

13. Conclusion

IncidentHawk meets its design goal of automating the critical first steps of Linux incident response: fast, read-only, dependency-free evidence collection paired with built-in threat heuristics and a defensible, hash-verified evidence bundle. Its modular collector design makes it straightforward to extend with new artifact types or detection rules as new attacker techniques emerge.

— End of Report —